Third System
AI Presence Index

The Presence Component Is Construct-Valid

A convergent–discriminant baseline: AI Presence tracks what it should and ignores what it should

June 2026 · Pablo Ulpiano González Castro

AIAS™ Measurement Program · Construct-Validity Baseline

AI Presence correlates strongly with what consumers search for (ρ = 0.74) and not at all with where products rank in retail sales (ρ = 0.00). A measure that behaves that way — close to the related signal, far from the unrelated one — is measuring a real, distinct thing. That is the textbook test of a valid construct, and the Presence component passes it.

ρ = **0.74** Convergent · ρ = **0.00** Discriminant · **0.74** Campbell–Fiske gap · **CONFIRMED** Baseline verdict

A measurement only earns the word "construct" when it is shown to track the things it should and stay clear of the things it should not. This baseline assembles two already-locked results into that test — it runs no new measurement of its own.

On the related side, AI Presence moves with consumer search interest: across 24 B2B SaaS brands, Presence and Google Trends correlate at ρ = 0.74 (v0.25). On the unrelated side, AI Presence is indifferent to retail sales rank: across 88 brands, Presence and Amazon Best Sellers Rank correlate at ρ = -0.0002 — statistically indistinguishable from zero (v0.26). The gap between the two, 0.74, is the single quantity this synthesis computes.

That pattern — strong where it should be strong, absent where it should be absent — is the Campbell–Fiske signature of a valid, distinct construct. The baseline verdict is CONFIRMED. The program proposes AI Availability as a third system alongside Mental and Physical Availability; for that proposal to stand, its Presence measure must first be shown valid — which is what this study establishes. The bounded finding is about the measure: Presence tracks brand salience and is distinct from retail sales rank, so it cannot be read off a search dashboard or a sales report. It has to be measured on its own terms.

What we measured

This is a synthesis. It performs no new acquisition; it reads two locked studies and computes one comparison.

The convergent leg (v0.25). AI Presence was correlated with Google Trends search interest across a 24-brand B2B SaaS panel. Search interest is a ***related*** signal — both reflect how present a brand is in the wider discourse — so a valid Presence measure should track it. It did: Spearman $\rho = 0.74$ ($p < 0.001$).

The discriminant leg (v0.26). AI Presence was tested against Amazon Best Sellers Rank across a pooled 88-brand set. Sales rank is an ***unrelated*** signal — it reflects price, distribution, and logistics, not AI salience — so a valid Presence measure should ***not*** track it. It did not: $\rho = -0.00$ ($p = 0.998$).

The test. Campbell & Fiske (1959) require both: correlation with the related trait, near-zero correlation with the unrelated one. The gap between the two coefficients — here 0.74 — is the evidence. A positive gap with the right significance pattern confirms the construct.

FINDING 01

Strong where it should be, absent where it should be

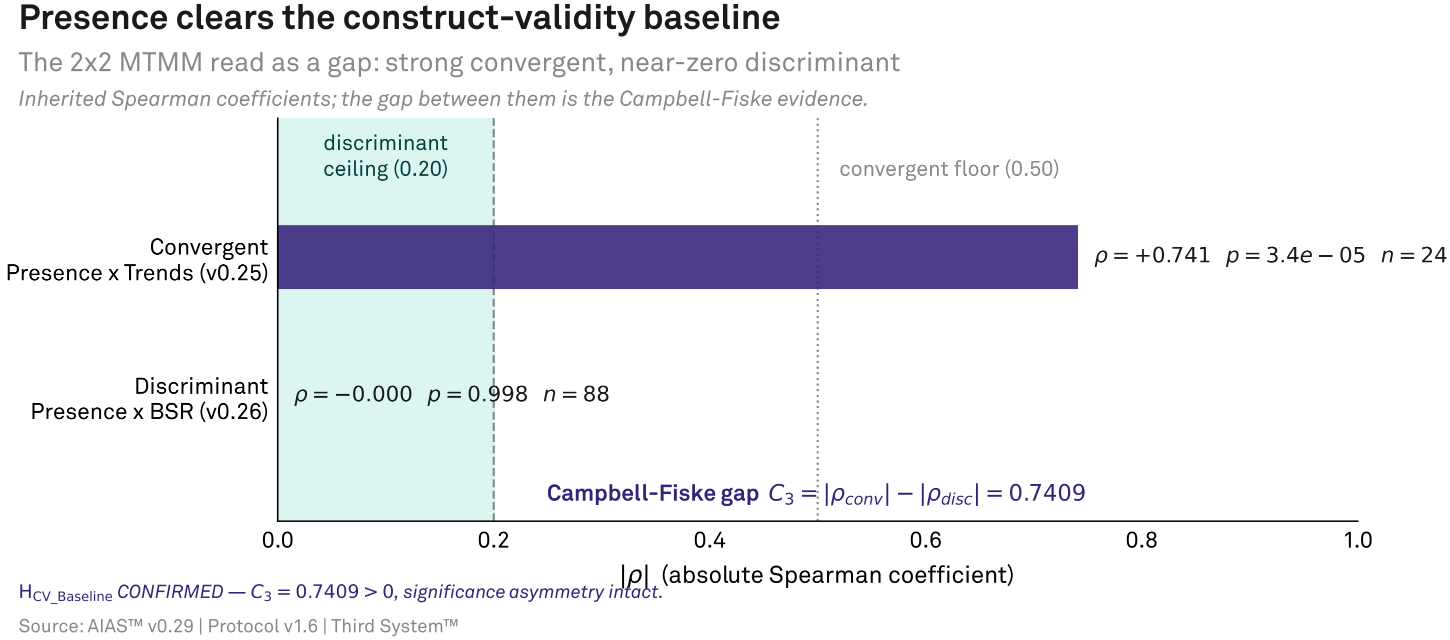


Figure 1 · The construct-validity gap. Convergent correlation magnitude (|ρ| = 0.74, significant) against discriminant magnitude (|ρ| = 0.0002, non-significant, inside the pre-registered 0.20 ceiling). The gap C_3 = 0.74 is the single quantity this synthesis computes; a positive gap with the significance asymmetry intact is the Campbell–Fiske evidence that the Presence component is construct-valid.

The whole case sits in one comparison. Against a related signal (search interest), AI Presence correlates at ρ = 0.74. Against an unrelated one (retail sales rank), it correlates at ρ = 0.00. The first is a strong, significant relationship; the second is statistically indistinguishable from zero.

A measure that correlated with both would be ambiguous — you could not say what it captured. A measure that correlated with neither would be noise. AI Presence does exactly what a valid, specific construct does: it lines up with the thing it is supposed to reflect and ignores the thing it is not.

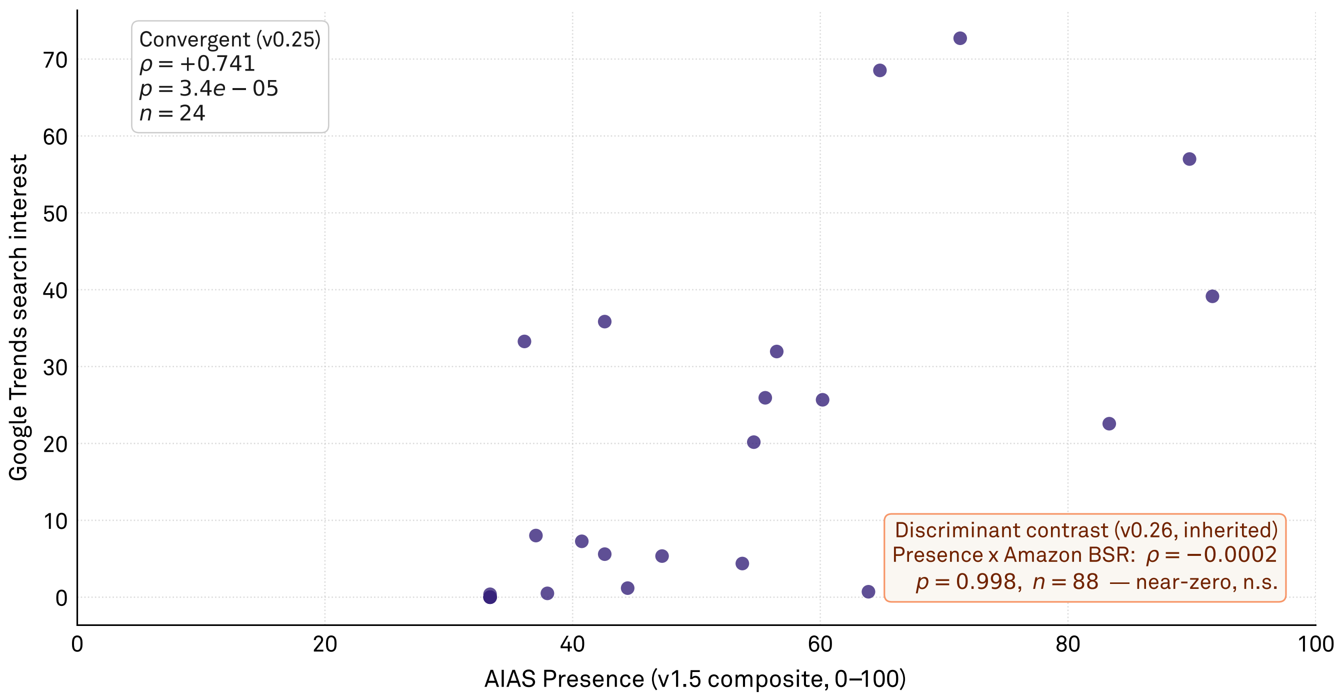
FINDING 02

The convergent leg: Presence tracks search demand

The convergent cell: Presence tracks search interest

v0.25 B2B SaaS — AIAS Presence x Google Trends

Per-point convergent data (left box). The discriminant cell is the inherited v0.26 contrast (lower right), asserted not re-plotted.



The two cells together are the Campbell-Fiske evidence: high convergent, near-zero discriminant.

Source: AIAS™ v0.29 | Protocol v1.6 | Third System™

Figure 2 · The convergent leg. AIAS Presence against Google Trends search interest across the 24-brand B2B SaaS registry (v0.25), Spearman ρ = 0.74. The discriminant relationship (Presence vs. Amazon Best Sellers Rank, ρ = -0.0002, n = 88) is the inherited v0.26 contrast, asserted not re-plotted — a synthesis reports its components' locked results rather than re-deriving them.

Across 24 B2B SaaS brands, the brands the models surface most are, by and large, the brands people search for most (ρ = 0.74, $p < 0.001$). This is the reassuring half of construct validity: AI Presence is anchored to a real-world salience signal, not floating free of it.

It matters that the related signal is search rather than sales. Search interest, like AI Presence, is a measure of mind-presence — how much a brand occupies the conversation — which is the dimension Presence is meant to capture. The strong tie to search is what makes the near-zero tie to sales meaningful rather than merely a weak measure.

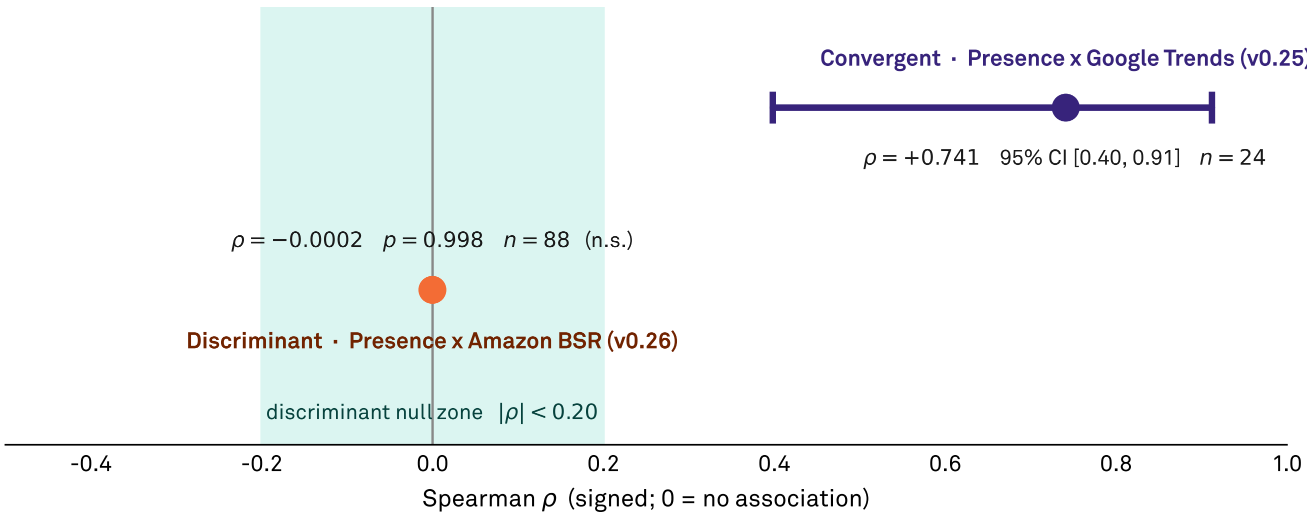
FINDING 03

The discriminant leg: Presence is not a sales proxy

Significant where expected, null where expected

The two legs on one signed-correlation axis

Convergent clears zero with its inherited 95% CI; discriminant sits inside the null zone, indistinguishable from zero.



The inferential asymmetry — one CI excludes zero, the other leg is n.s. — is the Campbell-Fiske signature.

Source: AIAS™ v0.29 | Protocol v1.6 | Third System™

Figure 3 · The discriminant leg, on a signed-correlation axis. The convergent coefficient (ρ = 0.74, v0.25) sits well above zero with its inherited 95% CI [0.40, 0.91] clearing both the null zone and zero; the discriminant coefficient (ρ = -0.0002, p = 0.998, n = 88, v0.26) sits inside the pre-registered |ρ| < 0.20 null zone, indistinguishable from zero. One CI excludes zero, the other leg is non-significant — the inferential asymmetry that makes the gap meaningful rather than a weak measure.

Against Amazon Best Sellers Rank — pooled across 88 brands — AI Presence correlates at ρ = -0.0002, p = 0.998. There is no relationship to detect. A brand can be highly present to the models and sell modestly, or sell heavily and be thin in the models; the two are independent.

This is not a weakness in the measure — it is the point. If AI Presence simply re-indexed sales, it would add nothing to the metrics brands already have. Its indifference to retail rank is what establishes it as a separate lever, worth measuring in its own right. The original v0.26 study framed this null as a failed prediction; in the construct-validity frame it is the evidence.

FINDING 04

Verdict: a construct-valid baseline for AI Availability

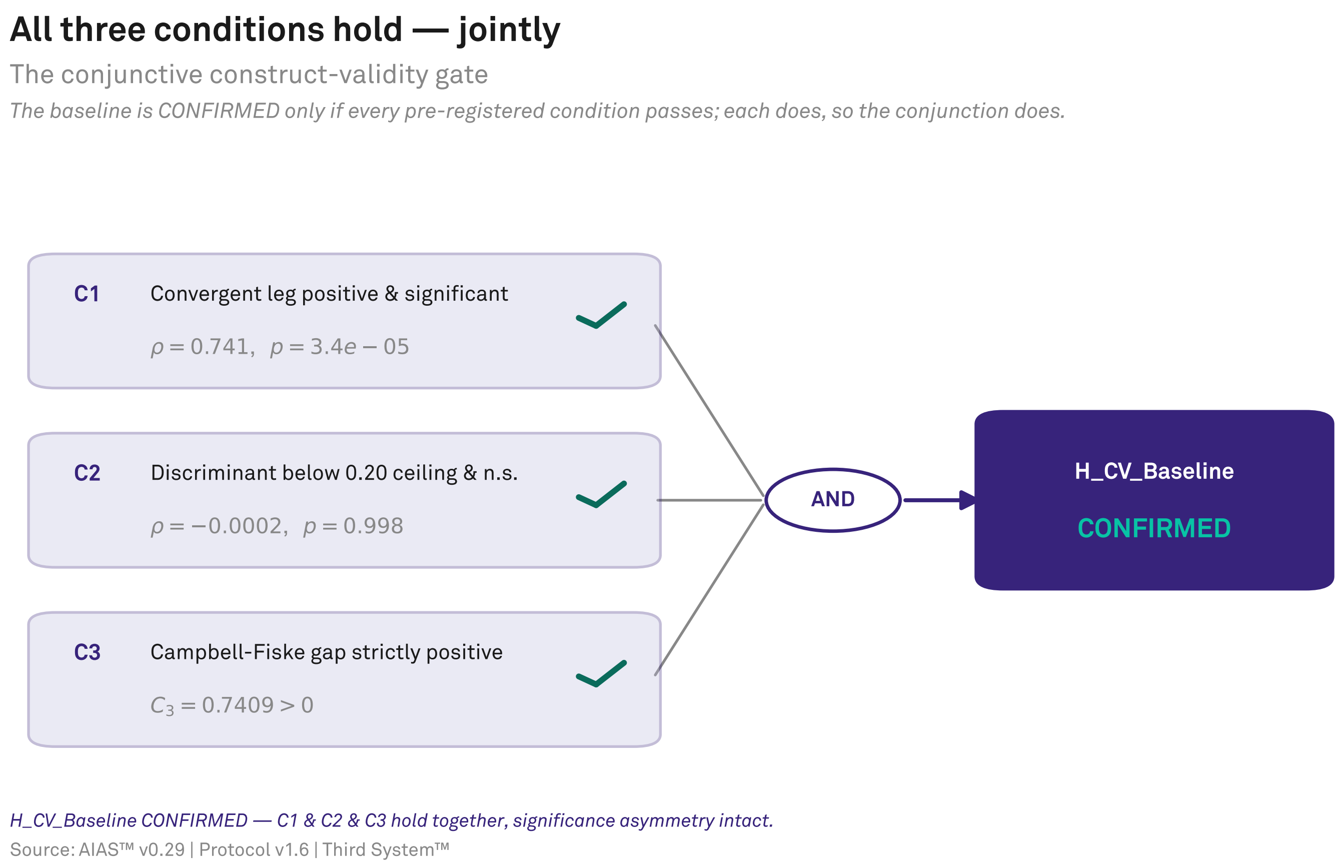


Figure 4 · The conjunctive verdict. Three pre-registered conditions — convergent positive and significant (C1), discriminant below the 0.20 ceiling and non-significant (C2), and a strictly positive gap (C3) — each hold, so their conjunction holds: H_CV_Baseline is CONFIRMED. The baseline is an AND of all three, not a majority; the gate makes that decision rule explicit.

Three pre-registered conditions had to hold together: the convergent leg positive and significant; the discriminant leg below a $|\rho| = 0.20$ ceiling and non-significant; and the gap between them strictly positive. All three hold, with the significance asymmetry intact — convergent significant, discriminant not. The baseline is CONFIRMED.

What this establishes is bounded and about the measure: the Presence component is construct-valid — measurable, anchored to a related salience signal, and distinct from commercial outcomes. That validity is the precondition the program's larger proposal depends on — that AI Availability is a third system alongside Mental and Physical Availability — but the proposal is the frame, not a result of this study. For brand leaders, the usable implication is narrower and firm: because Presence is valid and distinct, it cannot be inferred from search dashboards or sales reports; it has to be measured on its own surface.

Limitations

This is a baseline, not the last word. The convergent and discriminant legs come from different substrates (B2B SaaS for search; consumer goods for sales rank), so the synthesis assembles two contexts rather than one matched panel. A single panel measured against both signals at once would be a stronger MTMM cell.

The discriminant leg is a near-zero correlation, which is the desired result but also the easiest to obtain by accident; its weight comes from being paired with a strong, significant convergent leg, not from standing alone.

The predictive leg — whether AI Presence forecasts a future behavioral outcome — is not part of this baseline and is deferred to a later gated wave.

What’s next

The baseline establishes that Presence is construct-valid; the next questions are about reach. A matched-panel MTMM — one brand set measured against both a related and an unrelated signal simultaneously — would tighten the convergent–discriminant pair into a single design.

Beyond that, the predictive wave: testing whether AI Presence anticipates a downstream behavioral outcome would extend the construct from “distinct and well-anchored” to “decision-relevant.” That is the gate from a measurement claim to a management claim.

Hypothesis scoring

Four pre-registered conditions on the inherited convergent and discriminant verdicts; the baseline is CONFIRMED only if all hold jointly.

H	Pre-registered prediction	Result	Status
C1	Convergent leg positive and significant	$\rho = 0.7411$, $p = 3.4e\text{-}05$, $n = 24$ (inherited v0.25, H_CV_Primary).	**CONFIRMED**
C2	Discriminant leg below ceiling and non-significant	$\rho = -0.0002$, $p = 0.998$, $n = 88$ (inherited v0.26, H_PV_Pooled).	**CONFIRMED**
C3	Campbell–Fiske gap strictly positive	$C3 = \lvert 0.7411\rvert - \lvert -0.0002\rvert = 0.7409 > 0$; significance asymmetry intact.	**CONFIRMED**
H_CV_Baseline	Conjunctive construct-validity baseline	All three conditions hold jointly — the Presence component clears the baseline.	**CONFIRMED**

Detailed hypothesis results

C1 — Convergent leg: CONFIRMED. Inherited from v0.25 (SSRN 6842138, pre-reg v0.25-prereg-r1). AIAS Presence × Google Trends, 24-brand B2B SaaS panel: Spearman $\rho = 0.7411$, $p = 3.4e\text{-}05$, 95% CI [0.398, 0.912]. Positive and significant.

C2 — Discriminant leg: CONFIRMED. Inherited from v0.26 (SSRN 6847678, pre-reg v0.26-prereg-r2). AIAS Presence × Amazon Best Sellers Rank, pooled $n = 88$: $\rho = -0.0002$, $p = 0.998$ — below the $\lvert\rho\rvert = 0.20$ ceiling and non-significant.

C3 — Campbell–Fiske gap: CONFIRMED. $C3 = \lvert 0.7411\rvert - \lvert -0.0002\rvert = 0.7409 > 0$. The pre-registered point forecast was ~0.55; the realized gap exceeds it because the discriminant coefficient came in near zero rather than near the band ceiling. Forecast-versus-realization only; the lock is untouched.

H_CV_Baseline — Conjunctive verdict: CONFIRMED. C1 & C2 & C3 hold jointly with the significance asymmetry intact. The Presence component clears a convergent–discriminant construct-validity baseline.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.6). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

González Castro, P. U. (2026). ***The Presence Component Is Construct-Valid: A Convergent–Discriminant (Campbell–Fiske) Baseline for AI Availability (v0.29)***. Third System. thirdsystem.ai/v29 (SSRN 6870778)

Underlying datasets

Computed verdict: osf.io/ec6wh/v29/data/v29_verdicts.json

Convergent component (v0.25, SSRN 6842138): osf.io/ec6wh/v25

Discriminant component (v0.26, SSRN 6847678): osf.io/ec6wh/v26

Methodology log: v1.6 (SSRN 6816340).

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